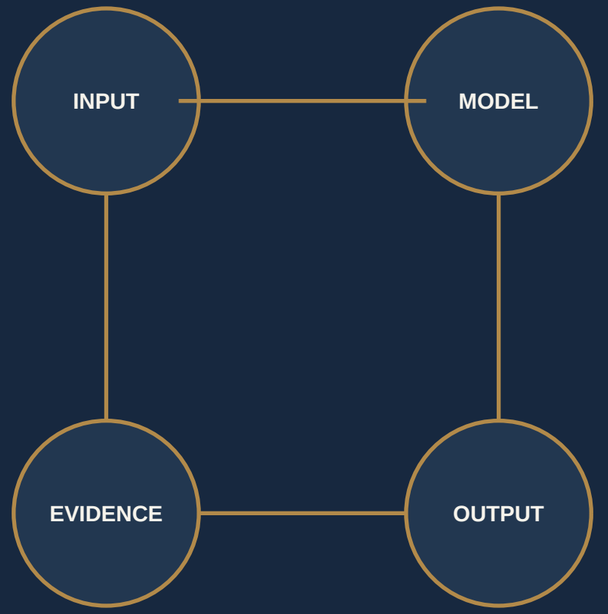
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Robot Localization and Mapping



AI.310 | University lecture series

What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define odometry.
- Apply key methods including probabilistic localization.
- Build or analyze a practical system involving kalman filters.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Odometry

02

METHODS

Probabilistic localization

03

SYSTEMS

Kalman filters

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Odometry

Odometry

02 · Probabilistic localization

Probabilistic localization

03 · Kalman filters

Kalman filters

04 · Particle filters

Particle filters

05 · Mapping and SLAM

Mapping and SLAM

06 · Mandatory practical: Implement

Mandatory practical: Implement localization and SLAM in a simulated robotic environment

Core ideas, made explicit

AI.310 • Distinguish definitions, assumptions and evidence

- Odometry
- Probabilistic localization
- Kalman filters
- Particle filters
- Mapping and SLAM

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.310 • Distinguish definitions, assumptions and evidence

- Mandatory practical: Implement localization and SLAM in a simulated robotic environment

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM

Odometry

INPUTS

Probabilistic localization

METHOD

Kalman filters

OUTPUT

Particle filters

EVALUATION

Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving odometry.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using odometry.
- Implement or simulate a baseline using probabilistic localization.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.